

WEEK-10--Elasticbeanstalk

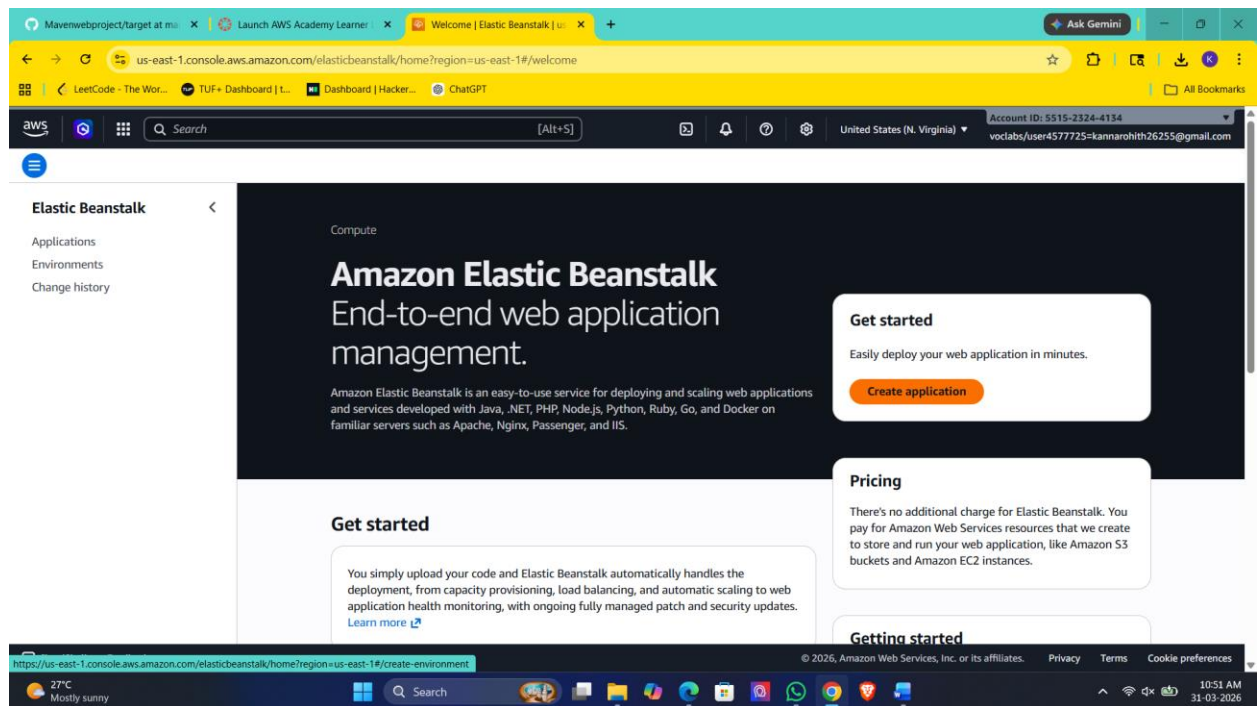
Prerequisites

- A web application ready (e.g., ROOT.war, .zip, or code)----**YOU CAN YOURS MAVEN PROJECT HERE**

Steps to Create Elastic Beanstalk Application

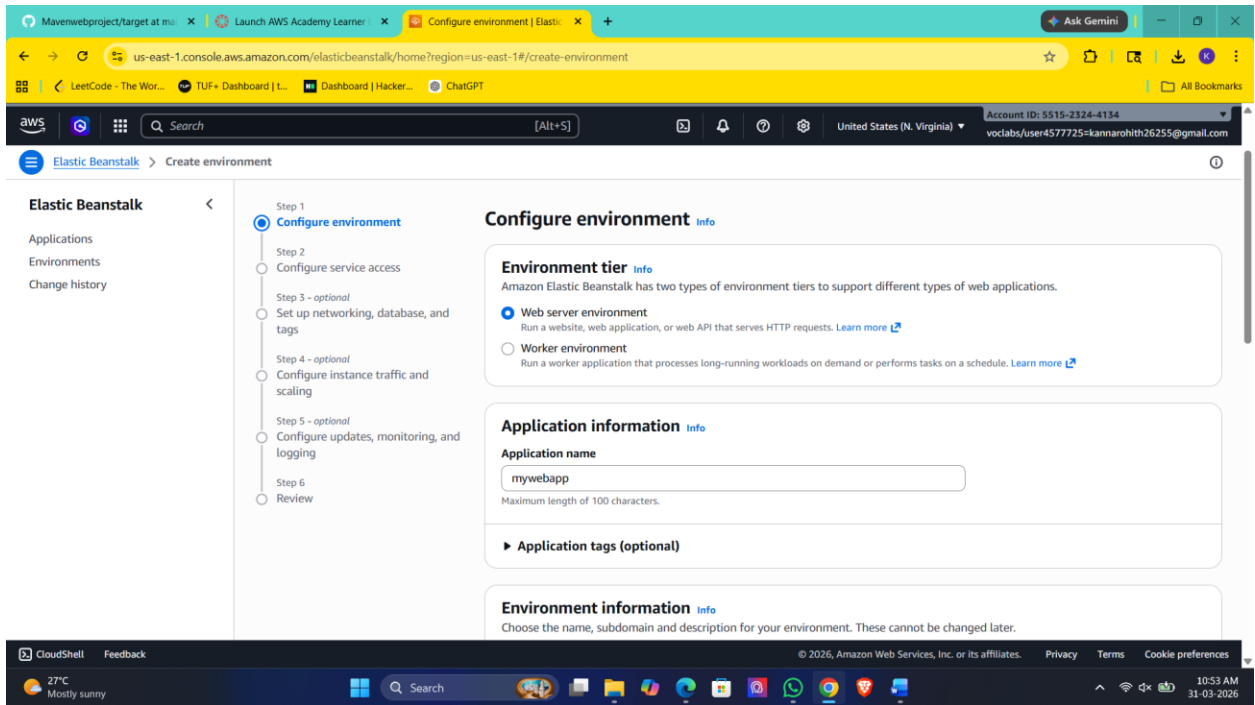
Step 1: Open Elastic Beanstalk

1. Login to AWS Console
2. Under Compute section ->Search for **Elastic Beanstalk**
3. Click **Create Application**



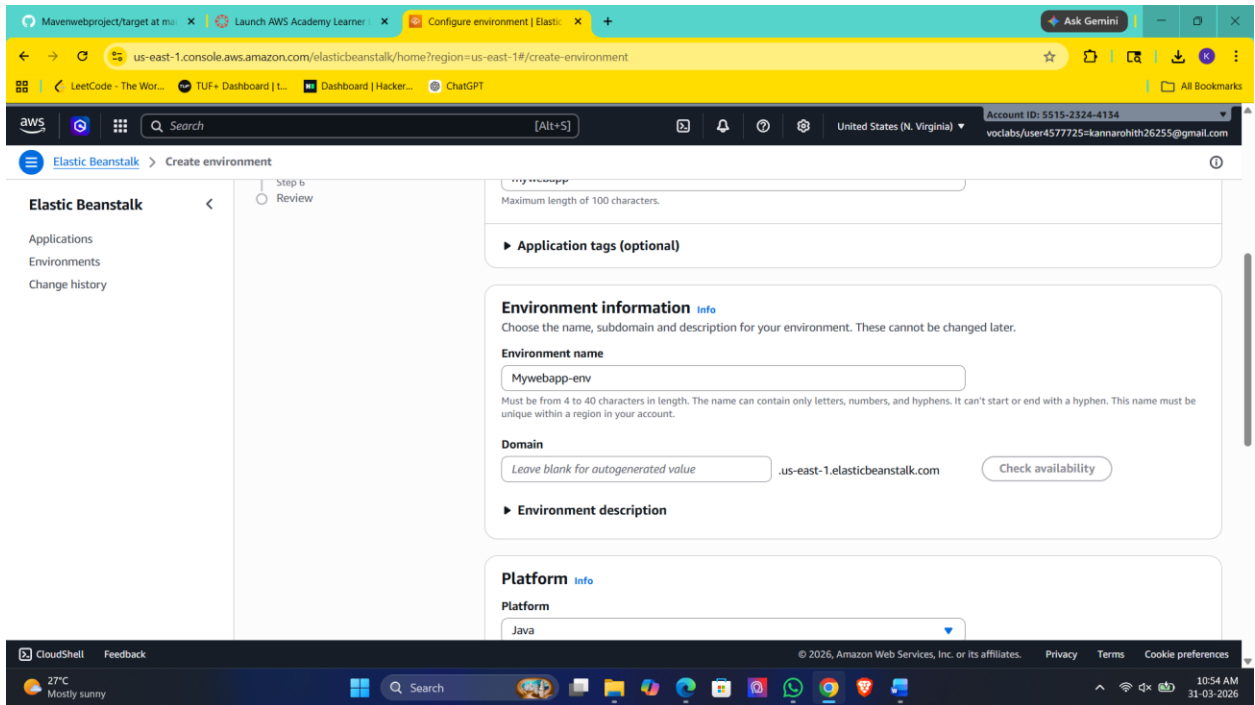
Step 2: Create Application

1. Enter **Application Name** (e.g., MyApp)
2. Add description (optional)
3. Click **Create**



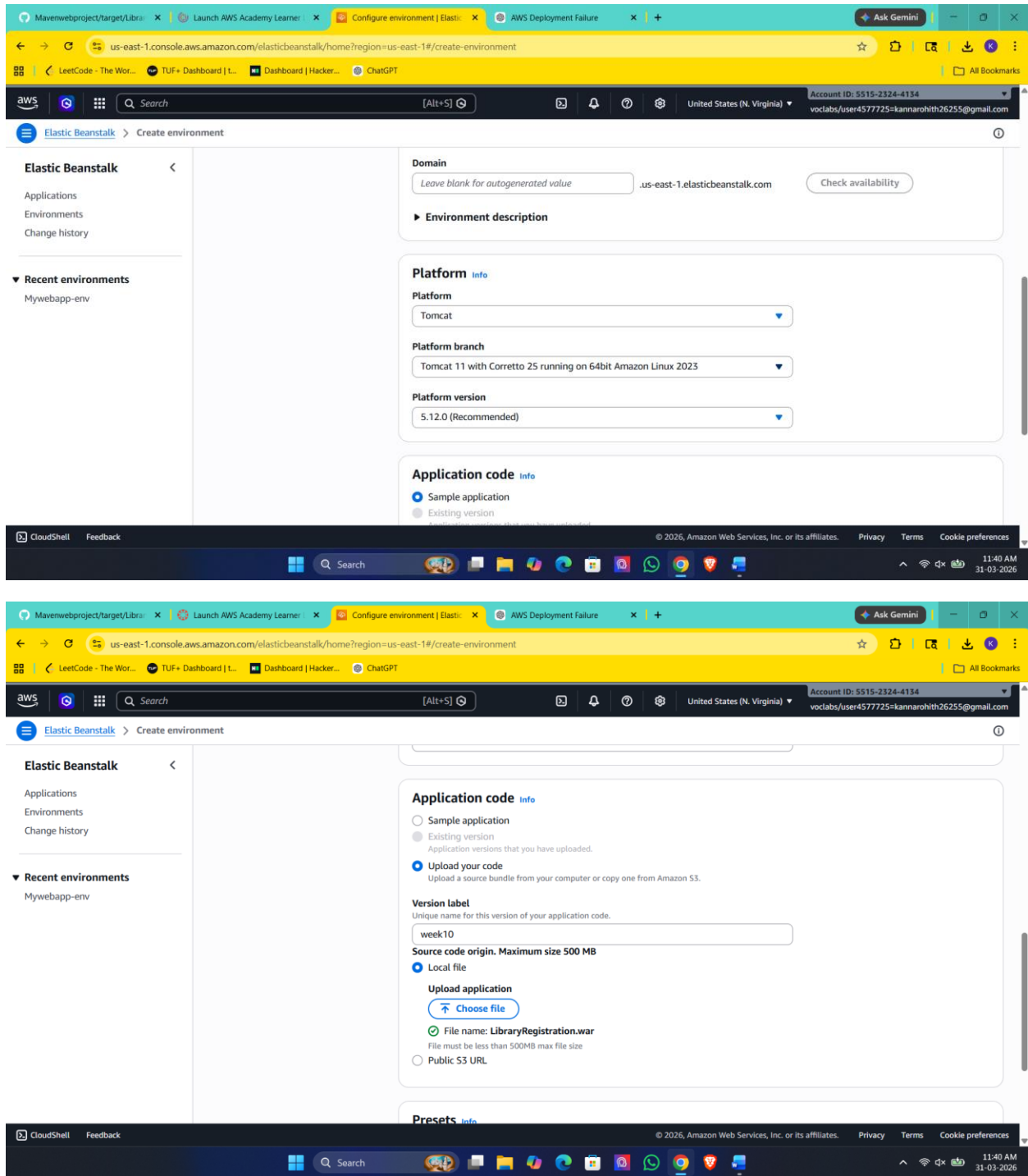
Step 3: Create Environment

1. Click **Create Environment**
2. Choose:
 - **Web Server Environment** (for web apps)
3. Click **Select**



Step 4: Configure Environment

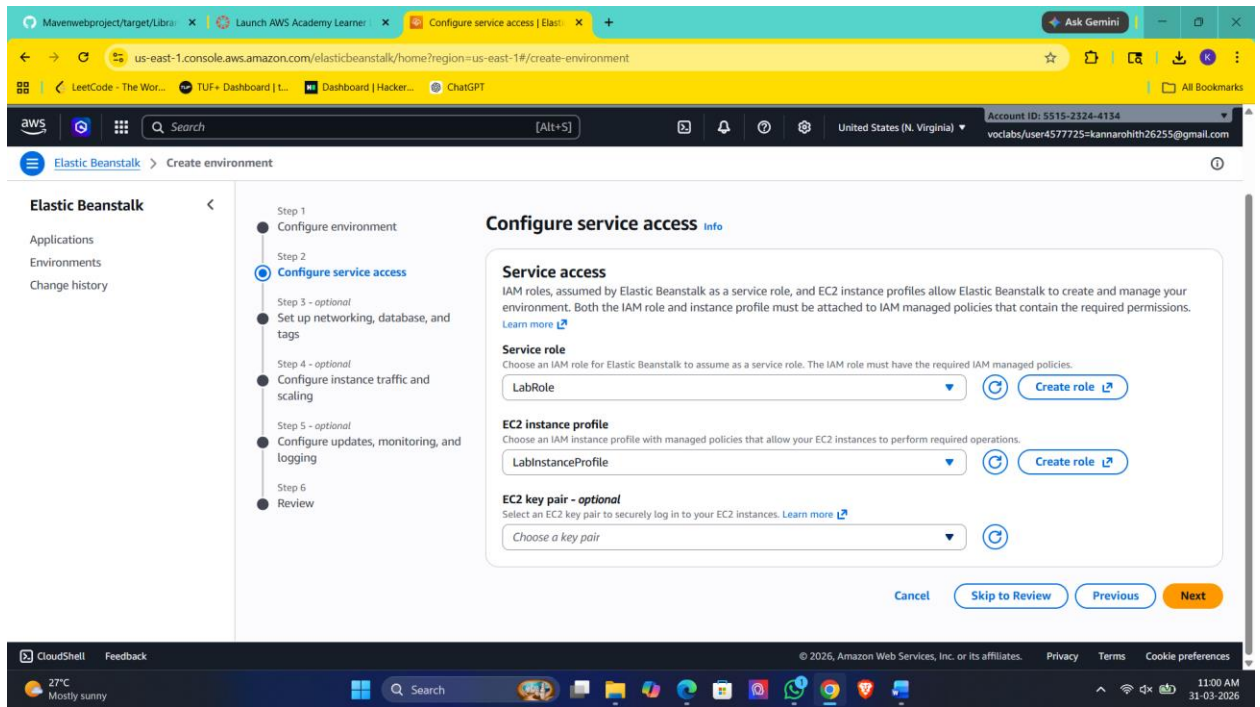
1. Enter **Environment Name**
2. Choose **Platform**:
 - Java (for .war)
 - Node.js / Python / PHP / etc.
3. Under **Application Code**:
 - Select **Upload your code**
 - Upload your .war or .zip file



Step 5: Configure Service Access

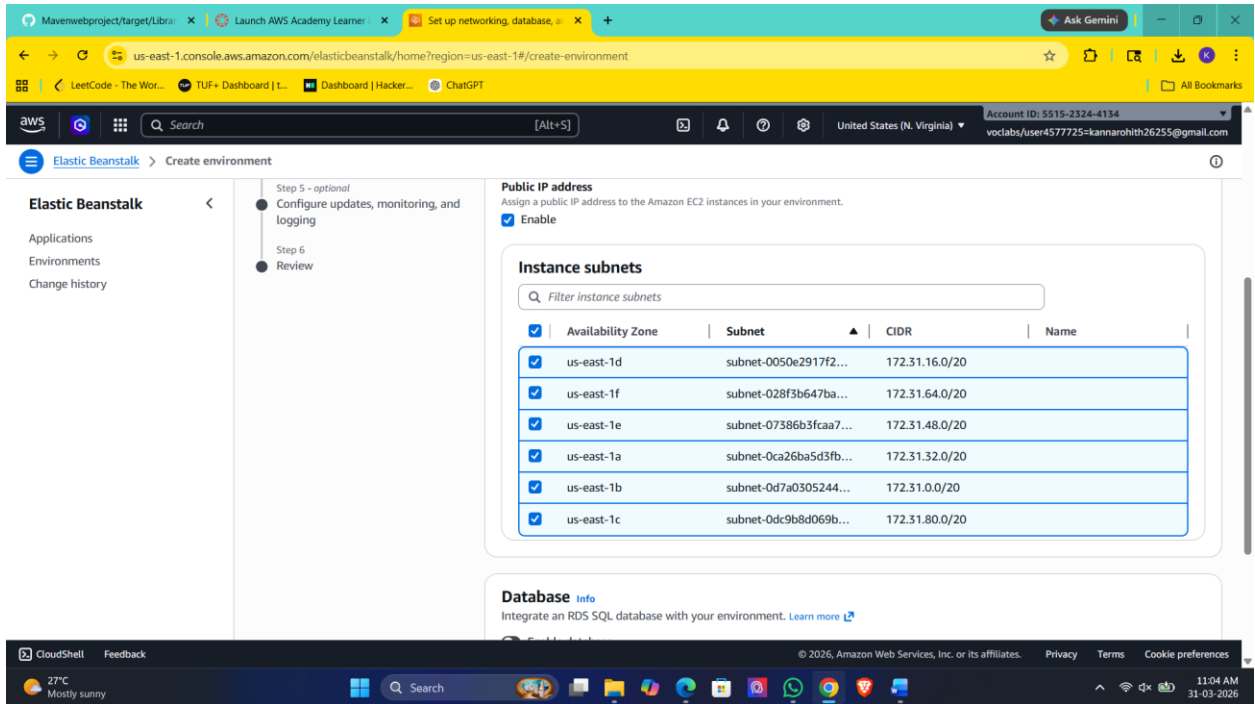
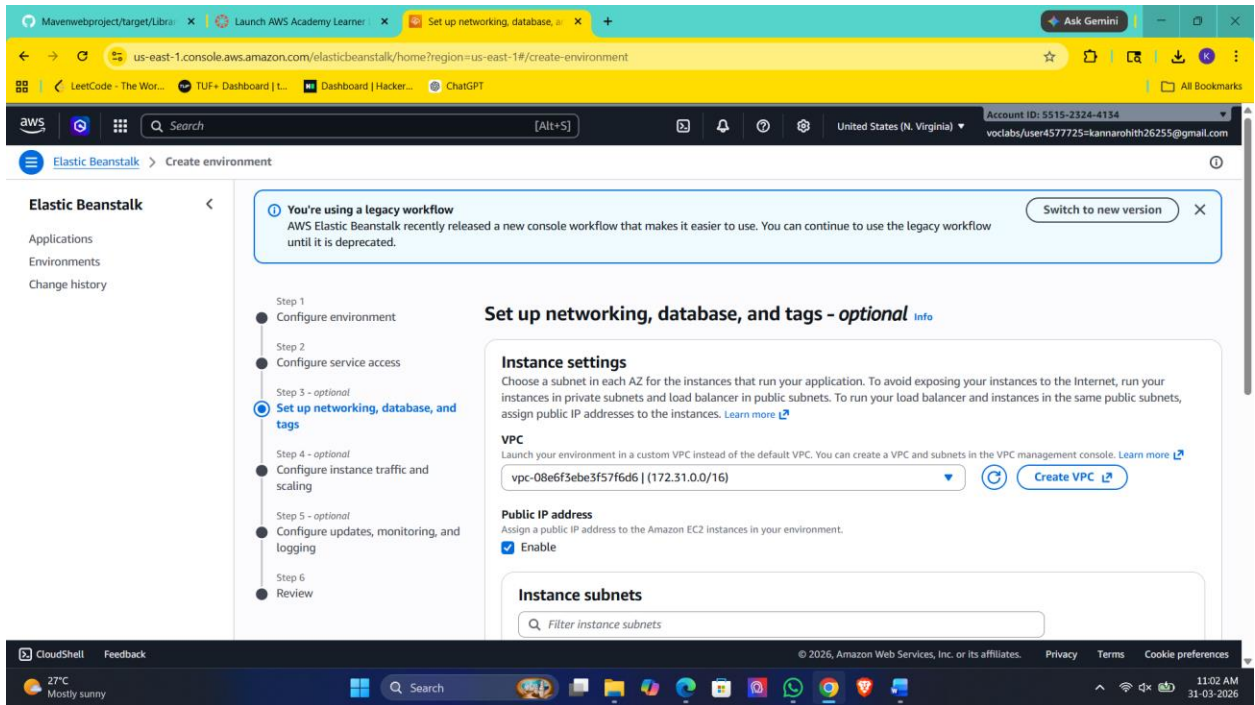
1. Service Role → Select existing → LabRole
2. EC2 Key Pair → Optional (for SSH)

3. EC2 Instance Profile → select (e.g., LabInstanceProfile)



Step 6: Configure Network

1. Select **VPC** (default is fine)
2. Select **Subnets**
3. Enable **Public IP**

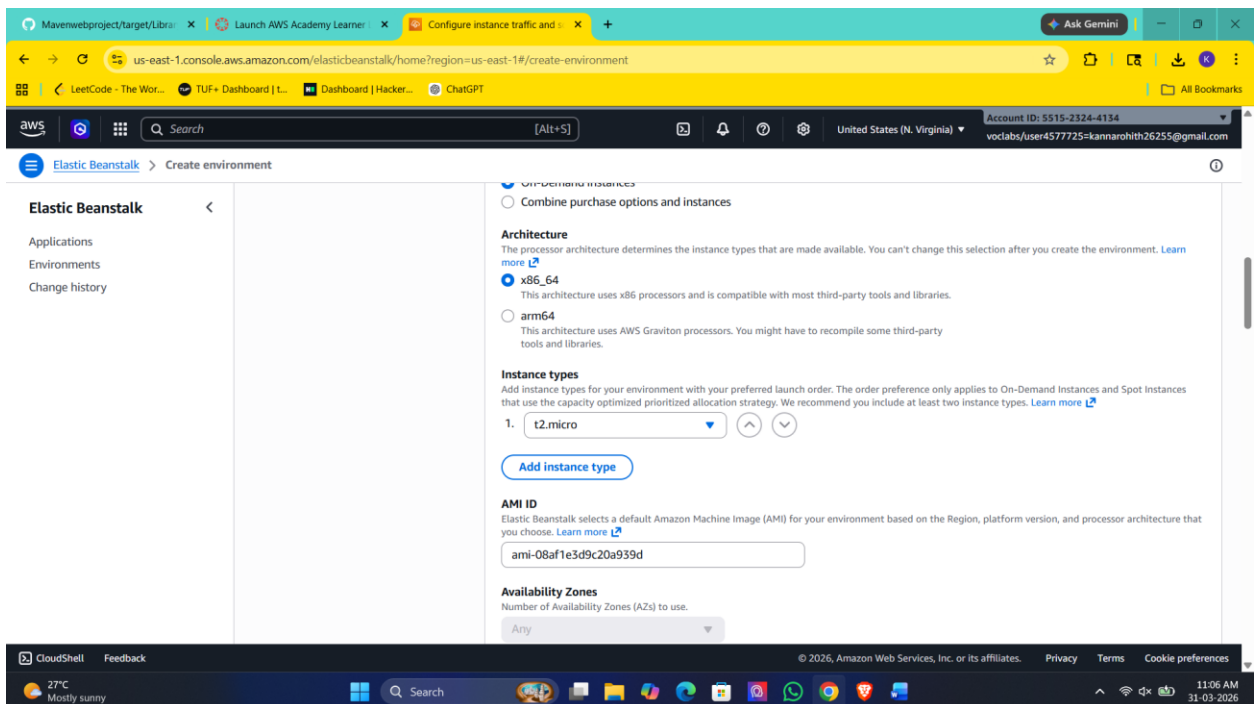
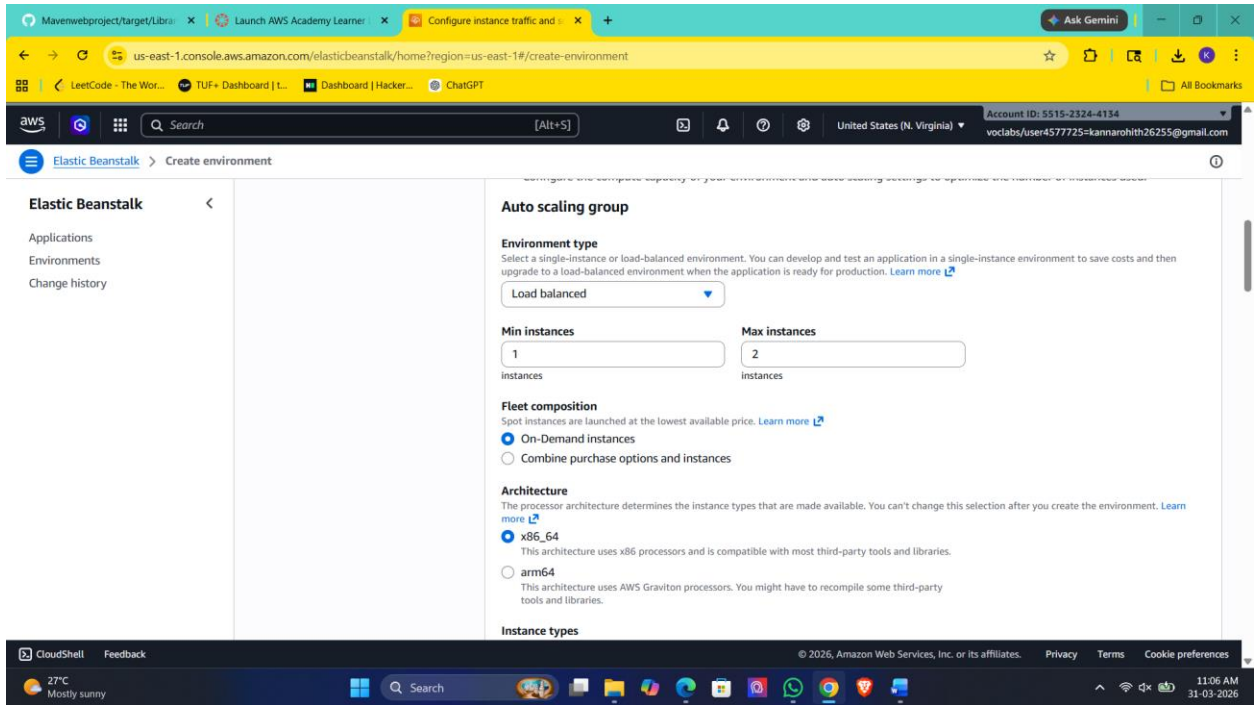


Step 7: Instance & Scaling Setup

1. Choose **Instance type** (e.g., t2.micro – free tier)
2. Set:

- Min instances: 1
- Max instances: 2 (or more)

3. Load Balancer → Enabled (optional but recommended)



Step 8: Review & Create

1. Click **Review**
2. Click **Create**

The screenshot shows the AWS Elastic Beanstalk console at the 'Create environment' page, specifically the 'Review' step. A notification banner at the top states: 'You're using a legacy workflow. AWS Elastic Beanstalk recently released a new console workflow that makes it easier to use. You can continue to use the legacy workflow until it is deprecated.' A 'Switch to new version' button is available. The left sidebar shows the navigation menu with 'Elastic Beanstalk' selected. The main content area is titled 'Review' and contains two sections: 'Step 1: Configure environment' and 'Step 2: Configure service access'. 'Step 1' is expanded, showing 'Environment information' with the following details:

Environment tier	Application name
Web server environment	mywebapp

Environment name	Application code
Mywebapp-env	LibraryRegistration.war

Platform: `arn:aws:elasticbeanstalk:us-east-1:platform/Corretto 25 running on 64bit Amazon Linux 2023/4.10.0`

'Step 2: Configure service access' is partially visible below. The bottom of the screen shows a Windows taskbar with the time 11:08 AM on 31-03-2026.

This screenshot shows the 'Service access' and 'Networking, database, and tags' steps of the AWS Elastic Beanstalk 'Create environment' process. The 'Service access' section is expanded, showing the 'Service role' as `arn:aws:iam::551523244134:role/LabRole` and the 'EC2 instance profile' as `LabInstanceProfile`. Below this, the 'Networking, database, and tags' section is expanded, showing 'Network' configuration:

VPC	Public IP address	Instance subnets
<code>vpc-08e6f3ebe3f57f6d6</code>	<code>true</code>	<code>subnet-0dc9b8d069b931153, subnet-0ca26ba5d3fb87451, subnet-0d7a03052448b6930, subnet-0050e2917f24e87fa, subnet-028f3b647ba6c845, subnet-07386b3fcaa765066</code>

The bottom of the screen shows a Windows taskbar with the time 11:09 AM on 31-03-2026.

Step 4: Configure instance traffic and scaling [Edit](#)

Instance traffic and scaling [info](#)
Customize the capacity and scaling for your environment's instances. Select security groups to control instance traffic. Configure the software that runs on your environment's instances by setting platform-specific options.

Instances

IMDSv1
Disabled

Capacity

Environment type	Min instances	Max instances
Load balanced	1	2

Fleet composition

On-Demand instances	On-demand base	On-demand above base
0	0	70

Capacity rebalancing

Disabled	Scaling cooldown	Processor type
360	x86_64	

Instance types

t2.micro	AMI ID	Availability Zones
ami-08af1e3d9c20a939d	Any	

Platform software

Log streaming	Logs retention	X-Ray enabled
Disabled	7	Disabled

Environment properties

Source	Key	Value
Plain text	GRADLE_HOME	/usr/local/gradle
Plain text	M2	/usr/local/apache-maven/bin
Plain text	M2_HOME	/usr/local/apache-maven

[Cancel](#) [Previous](#) [Create](#)

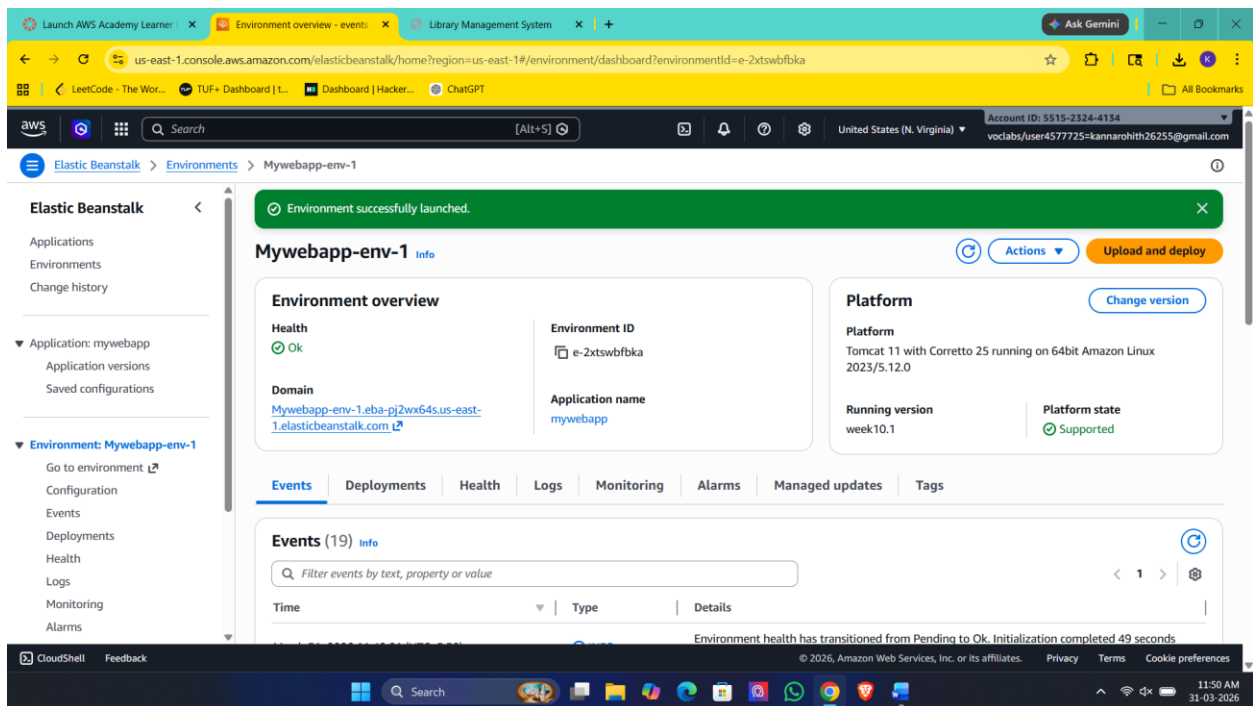
Deployment Process

- AWS automatically creates:
 - EC2 instances

- Load Balancer
- Auto Scaling Group
- Security Groups

Access Your Application

- After deployment:
 - You will get a **URL (domain link)**
 - Open it in browser → Your app will run





Welcome to the Library Management System

This is a simple web application built using JSP and Maven.

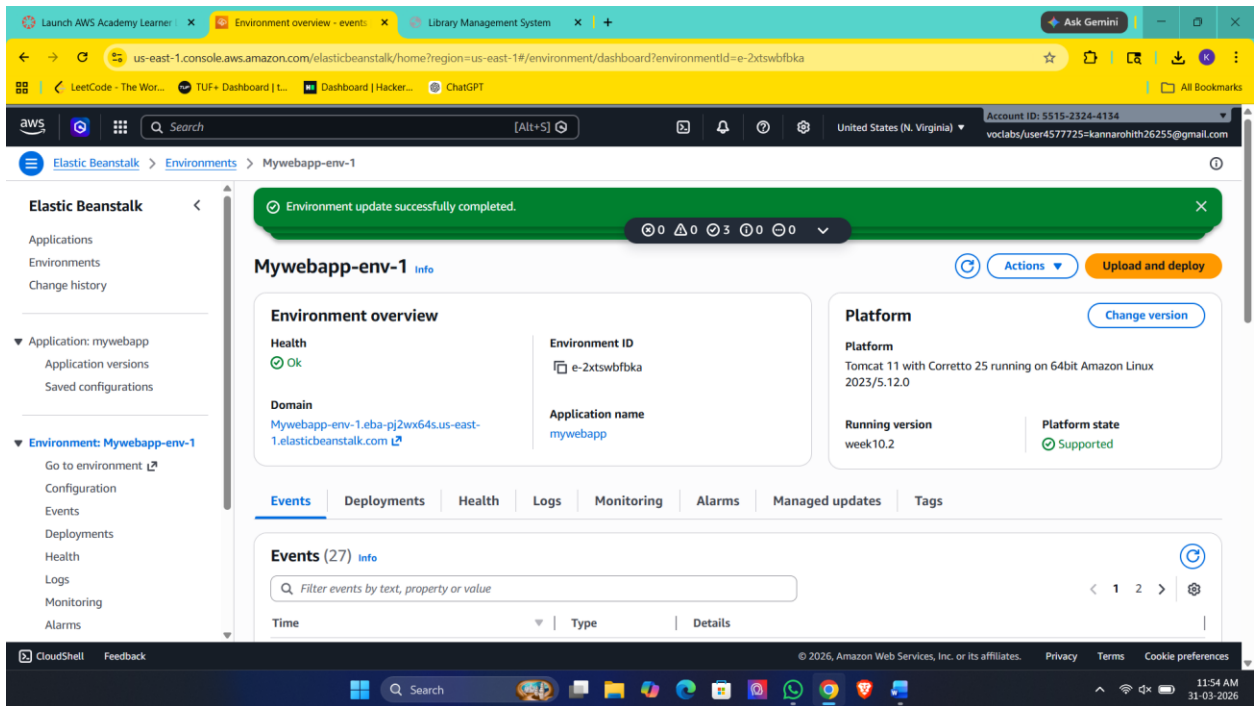
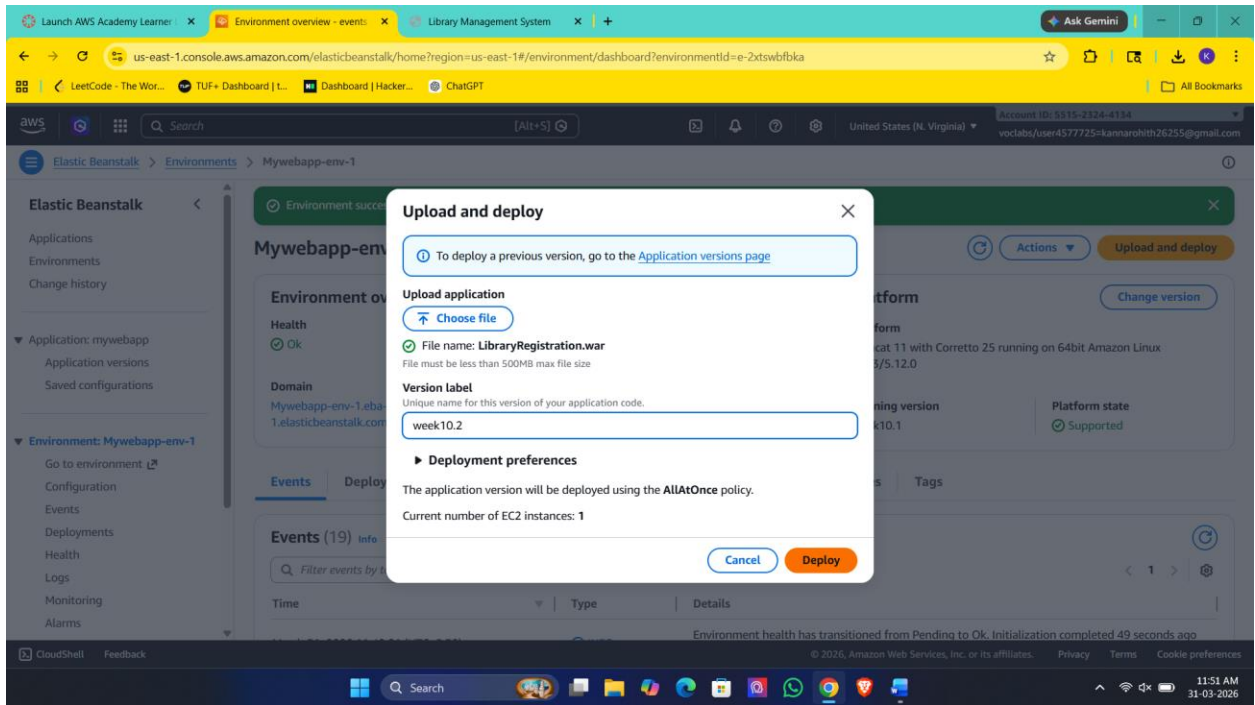
What do you want to do?

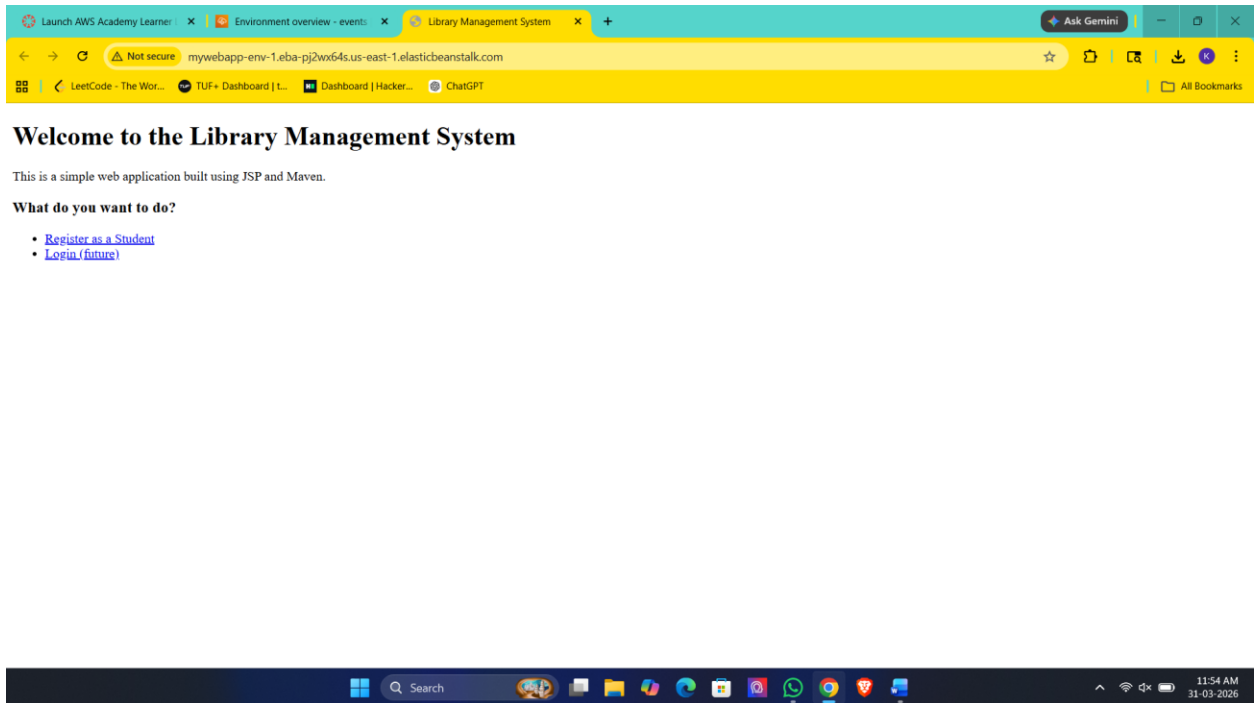
- [Register as a Student](#)
- [Login \(future\)](#)



Updating Application

1. Go to your environment
2. Click **Upload and Deploy**
3. Upload new version





Scenario-Based Questions

Scenario 1: Application Not Loading

Q1. What could be the possible reasons?

- Application crashed or not started
- Wrong port or missing entry file

Q2. Could deployment or configuration be incorrect?

Yes. Wrong platform or incorrect packaging can prevent the app from running.

Q3. How would you troubleshoot this problem?

Check **Events** for failure messages and logs like eb-engine.log to identify startup errors.

Scenario 2: Incorrect Application Structure

Q1. How does application structure affect deployment?

Elastic Beanstalk requires files at the **root level**; incorrect structure prevents detection of the app.

Q2. What happens if files are inside an extra folder?

EB cannot find the entry file, causing deployment failure or blank output.

Q3. How would you fix the structure issue?

Move required files (e.g., .war, app.js) to the root and repackage the ZIP correctly.

Scenario 3: Wrong Platform Selection

Q1. Why is platform selection important?

Platform defines runtime (Java, Node, Python) and how the app is executed.

Q2. What errors might occur?

App won't start or shows runtime errors due to mismatch.

Q3. How can you correct the platform?

Recreate the environment with the correct platform (e.g., Tomcat for .war).

Scenario 4: Environment Health Warning

Q1. Possible reasons for warning?

- Failed health checks
- High response time or errors

Q2. Which sections to check first?

Events, Logs, and Metrics.

Q3. How to resolve?

Fix health check URL, reduce errors, and optimize performance.

Scenario 5: Application Errors in Logs

Q1. What do such errors indicate?

The application is failing during execution.

Q2. Could misconfiguration cause this?

Yes, missing environment variables or resources can break the app.

Q3. How to analyze logs?

Identify the first error, read stack trace, and trace the failing component.

Scenario 6: Deployment Failure

Q1. What could be wrong with the package?

Missing entry file or incorrect ZIP structure.

Q2. Which logs to check?

eb-engine.log and Events tab.

Q3. How do packaging errors affect deployment?

EB cannot start the app, leading to complete failure.

Scenario 7: Works Locally but Not on Cloud**Q1. What environment differences cause this?**

OS differences, ports, or file paths.

Q2. Could dependencies/config be responsible?

Yes, missing dependencies or incorrect configs.

Q3. How to debug?

Compare local vs cloud setup and analyze logs.

Scenario 8: Blank Output**Q1. What causes no output?**

Routes not defined or frontend not loading.

Q2. Could runtime/config issues be responsible?

Yes, incorrect responses or missing files.

Q3. How do logs help?

Logs show request handling and response errors.

Scenario 9: Application URL Not Accessible**Q1. Could environment still be launching?**

Yes, during initialization phase.

Q2. What status indicators to check?

Environment status and health (Pending, Green).

Q3. How can Events help?

They show deployment progress and failures.

Scenario 10: Resource Management Issue**Q1. Cost implications?**

Charges continue for running resources.

Q2. Which services run in background?

EC2, Load Balancer, Auto Scaling.

Q3. How to terminate resources?

Delete environment and verify all services are stopped.

Scenario 11: High Traffic Handling**Q1. How does EB handle scaling?**

Using Auto Scaling Groups.

Q2. What configurations are needed?

Scaling policies and load balancer.

Q3. What if scaling is not configured?

App slows down or crashes under load.

Scenario 12: Application Update Not Reflected**Q1. Could deployment issues be the reason?**

Yes, old version may still be active.

Q2. How can caching affect output?

Browser or CDN may serve old content.

Q3. How to ensure latest version?

Redeploy properly and clear cache.